InsightEngine: An End-to-End Data Engineering Platform for Business Intelligence

Team Members :

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Overview

Abstract

InsightEngine is an end-to-end data engineering platform designed to collect, ingest, process, transform, store, and analyze data from multiple business sources to support data-driven decision-making. The platform simulates real-world data engineering workflows by generating structured and semi-structured business data using Python and ingesting it through batch and streaming mechanisms. Apache Kafka is used for data ingestion, PostgreSQL is used for operational and analytical storage, and PySpark is used for scalable data processing and transformations. Apache Airflow orchestrates and automates the complete data pipeline, including data validation, cleaning, transformation, and loading into a structured data warehouse based on dimensional modeling principles. The processed data is presented through an interactive Streamlit dashboard to provide meaningful business insights and create a helpfull dashboard which changes with data time to time. The project demonstrates key data engineering concepts including source systems, data architecture, storage systems, ingestion strategies, batch processing, data modeling, transformations, and the complete data engineering lifecycle.

Introduction

In today's data-driven environment, organizations generate large volumes of data from multiple sources such as customer transactions, products, payments, inventory, and business operations. Raw data from these sources is often distributed, inconsistent, and difficult to use directly for decision-making. Data engineering provides the processes and technologies required to collect, ingest, store, process, transform, and organize this data into reliable and meaningful information.

InsightEngine is an end-to-end data engineering platform designed to demonstrate the complete data engineering lifecycle. The platform generates realistic business data using Python, supports batch and streaming data ingestion through Python and Apache Kafka, stores operational data in PostgreSQL, and performs data cleaning and transformation using PySpark. Apache Airflow is used to orchestrate and automate the pipeline, while the transformed data is stored in a structured data warehouse using dimensional modeling principles. Finally, an interactive Streamlit dashboard presents business insights. The project integrates key data engineering concepts including source systems, data architecture, storage, ingestion, batch processing, data modeling, transformations, and data quality.

Objectives

- **To develop an integrated platform that demonstrates the complete data engineering lifecycle.**
- **To connect the entire data pipeline with a simple interface for consuming business data and insights.**
- **To demonstrate how an integrated data engineering workflow supports data-driven decision-making.**

Literature Survey

Paper Ref	Authors	Drawbacks	Future Work	Link
The dataflow model	T. Akidau et al.	•Handling late and out-of-order data is complex. •Balancing **accuracy, latency, and cost** is challenging. •Requires significant computational resources for large-scale data.	•Improve processing efficiency and scalability. •Better handling of late/out-of-order data. •Explore more efficient resource management.	DOI / PAPER
Kafka: a Distributed Messaging System for Log Processing	J. Kreps, N. Narkhede, J. Rao	•Limited message durability in the original design. •Does not provide strong guarantees for all message delivery. •Basic Kafka focuses on messaging rather than stream processing.	•Add **replication** for better durability and availability. •Add **stream-processing capabilities** such as window-based counting and joins.	Paper / PDF
Apache Spark: a unified engine for big data processing	M. Zaharia et al	•High memory usage for large-scale data processing. •Performance can decrease for very large or complex workloads. •Requires cluster resources and proper configuration.	•Improve Spark's **performance and resource efficiency**. •Better support for **large-scale and real-time processing**.	DOI / Paper
Dremel: interactive analysis of web-scale datasets	S. Melnik et al.	•Mainly designed for **read-only data**, limiting dynamic data updates. •Best suited for **analytical queries**, not general-purpose data processing.	•Support more **dynamic and frequently changing data**. •Improve query performance and resource efficiency.	DOI / Paper

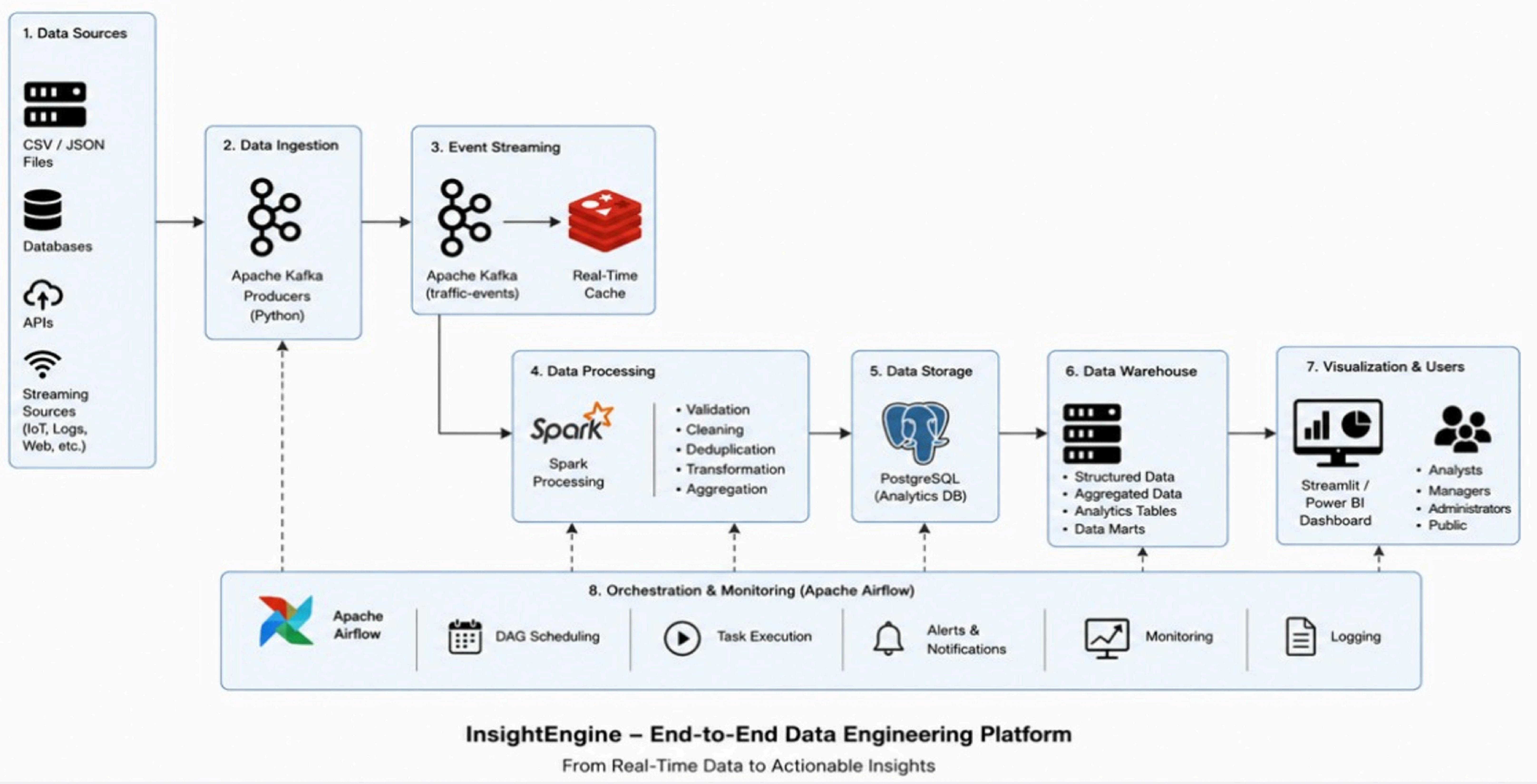
Literature Survey

Paper Ref	Authors	Drawbacks	Future Work	Link
A performance modeling framework for lambda architecture based applications	M. Gribaudo, M. Iacono, M. Kiran	•Results depend on **workload and system configuration**. •May not fully represent the behaviuor of real-world applications.	Improve the **accuracy of performance models**.	DOI / Paper
Data Lakes: A Survey of Functions and Systems	R. Hai et al.	•Data lakes can become **data swamps** without proper metadata and management. •Managing data from **different sources and formats** is challenging.	Improve metadata management and integration	IEEE / DOI
Data pipeline quality: Influencing factors, root causes of data-related issues, and processing problem areas for developers	H. Foidl et al.	Data pipelines face **data quality and integration issues**.	Improve automated data-quality mechanisms	DOI / Paper
An empirical study of developers' challenges in implementing Workflows as Code: A case study on Apache Airflow	J. Yasmin et al.	•Developers face **complex workflow configuration and execution issues**. •Requires **Airflow knowledge and proper environment setup**.	•Improve **Airflow documentation and developer support**. •Develop tools to **detect configuration and workflow errors automatically**.	DOI / Paper

Literature Survey

Paper Ref	Authors	Drawbacks	Future Work	Link
A proposed model for data warehouse ETL processes	A. Simitsis et al.	ETL development is complex and resource-intensive	Improve automated ETL design and optimization	DOI / Paper
Generative AI and Agentic Orchestration for Autonomous Data Engineering in Multi-Domain Enterprise Analytics Platforms	Mallikarjuna Rao Vasa	Focuses heavily on GenAI/agentic automation; requires complex multi-agent architecture	Extend autonomous ETL, governance and multi-domain data engineering	DOI / Paper

System Architecture



Thank You